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Assignment : 1-5;

Assignment –1

Creating sample tables and inserting values.

Create the following tables with the given structures and insert sample data as specified: -

A) SALESPEOPLE

|       |             |
|-------|-------------|
| Snum  | int(4)      |
| Sname | varchar(10) |
| City  | varchar(10) |
| Comm  | float(3,2)  |

B) CUSTOMERS

|        |             |
|--------|-------------|
| Cnum   | int(4)      |
| Cname  | varchar(10) |
| City   | varchar(10) |
| Rating | int(4)      |
| Snum   | int(4)      |

C) ORDERS

|       |            |
|-------|------------|
| Onum  | int(4)     |
| Amt   | float(7,2) |
| Odate | date       |
| Cnum  | int(4)     |
| Snum  | int(4)     |

```
mysql> create table SALESPEOPLE(Snum int(4),Sname varchar(10),City varchar(10),Comm float(3,2));
```

Query OK, 0 rows affected, 2 warnings (0.42 sec)

```
mysql> create table CUSTOMERS(
```

```
-> Cnum int(4),
```

```
-> Cname varchar(10),
```

```
-> City varchar(10),
```

```
-> Rating int(4),
```

```
-> Snum int(4));
```

Query OK, 0 rows affected, 3 warnings (0.05 sec)

```
mysql> create table ORDERS(
```

-> Onum int(4),  
 -> Amt float(7,2),  
 -> Odate date,  
 -> Cnum int(4),  
 -> Snum int(4));

Query OK, 0 rows affected, 4 warnings (0.04 sec)

```
mysql> show tables;
+-----+
| Tables_in_cdacmumbai |
+-----+
| customers             |
| orders                |
| salespeople           |
+-----+
3 rows in set (0.18 sec)

mysql>
      show full tables;
+-----+-----+
| Tables_in_cdacmumbai | Table_type |
+-----+-----+
| customers             | BASE TABLE |
| orders                | BASE TABLE |
| salespeople           | BASE TABLE |
+-----+-----+
3 rows in set (0.00 sec)

mysql> desc customers
-> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Cnum  | int           | YES  |     | NULL    |       |
| Cname | varchar(10)   | YES  |     | NULL    |       |
| City  | varchar(10)   | YES  |     | NULL    |       |
| Rating | int           | YES  |     | NULL    |       |
| Snum  | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.07 sec)
```

### SALES PEOPLE

| SNUM | SNAME   | CITY      | COMM |
|------|---------|-----------|------|
| 1001 | Peel    | London    | .12  |
| 1002 | Serres  | San Jose  | .13  |
| 1004 | Motika  | London    | .11  |
| 1007 | Rifkin  | Barcelona | .15  |
| 1003 | Axelrod | New York  | .10  |

```
insert into Salespeople values (1001,'Peel','London',.12),(1002,'Serres','San Jose',.13),
(1004,'Motika','London',.12), (1007,'Rifkin','Barcelona',.15),
(1003,'Axelord','New York',.10);
```

### CUSTOMERS

| CNUM | CNAME    | CITY     | RATING | SNUM |
|------|----------|----------|--------|------|
| 2001 | Hoffman  | London   | 100    | 1001 |
| 2002 | Giovanni | Rome     | 200    | 1003 |
| 2003 | Liu      | San Jose | 200    | 1002 |
| 2004 | Grass    | Berlin   | 300    | 1002 |
| 2006 | Clemens  | London   | 100    | 1001 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
| 2007 | Pereira  | Rome     | 100    | 1004 |

```
mysql> insert into Customers values(2001,'Hoffman','London',100,1001), (2002 , 'Giovanni' ,
'Rome' , 200 , 1003),(2003 , 'Liu' , 'San Jose' , 200 , 1002),(2004 , 'Grass' , 'Berlin' , 300 ,
1002), (2006 , 'Clemens' , 'London' , 100 , 1001),(2008 , 'Cisneros' , 'San Jose' , 300 , 1
007),(2007 , 'Pereira' , 'Rome' , 100 , 1004);
```

### ORDERS

| ONUM | AMT     | ODATE         | CNUM | SNUM |
|------|---------|---------------|------|------|
| 3001 | 18.69   | 03-OCT-1990   | 2008 | 1007 |
| 3003 | 767.19  | 03- OCT -1990 | 2001 | 1001 |
| 3002 | 1900.10 | 03- OCT -1990 | 2007 | 1004 |
| 3005 | 5160.45 | 03- OCT -1990 | 2003 | 1002 |
| 3006 | 1098.16 | 03- OCT -1990 | 2008 | 1007 |
| 3009 | 1713.23 | 04- OCT -1990 | 2002 | 1003 |
| 3007 | 75.75   | 04- OCT -1990 | 2004 | 1002 |
| 3008 | 4723.00 | 05- OCT -1990 | 2006 | 1001 |
| 3010 | 1309.95 | 06- OCT -1990 | 2004 | 1002 |
| 3011 | 9891.88 | 06- OCT -1990 | 2006 | 1001 |

```
mysql> insert into Orders values(3001 , 18.69 , '1990-10-03' , 2008 , 1007);
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into Orders values(3003 , 767.19 , '1990-10-03' , 2001 , 1001);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into Orders values(3002 , 1900.10 , '1990-10-03' , 2007 , 1004);
```

Query OK, 1 row affected (0.01 sec)

mysql> insert into Orders values(3005 , 5160.45,'1990-10-03', 2003 , 1002);

Query OK, 1 row affected (0.01 sec)

mysql> insert into Orders values(3006 ,1098.16,'1990-10-03',2008 , 1007);

Query OK, 1 row affected (0.00 sec)

mysql> insert into Orders values(3009,1713.23,'1990-10-04',2002 , 1003);

Query OK, 1 row affected (0.01 sec)

mysql> insert into Orders values(3007,75.75,'1990-10-04',2004,1002);

Query OK, 1 row affected (0.01 sec)

mysql> insert into Orders values(3008,4723.00,'1990-10-05',2006,1001);

Query OK, 1 row affected (0.00 sec)

mysql> insert into Orders values(3010 ,1309.95 ,'1990-10-06',2004,1002);

Query OK, 1 row affected (0.00 sec)

mysql> insert into Orders values(3011,9891.88,'1990-10-06',2006,1001);

Query OK, 1 row affected (0.01 sec)

```
mysql> select * from Salespeople;
+-----+-----+-----+-----+
| Snum | Sname | City | Comm |
+-----+-----+-----+-----+
| 1001 | Peel | London | 0.12 |
| 1002 | Serres | San Jose | 0.13 |
| 1004 | Motika | London | 0.12 |
| 1007 | Rifkin | Barcelona | 0.15 |
| 1003 | Axelord | New York | 0.10 |
+-----+-----+-----+-----+
5 rows in set (0.02 sec)

mysql> select * from customers;
+-----+-----+-----+-----+-----+
| Cnum | Cname | City | Rating | Snum |
+-----+-----+-----+-----+-----+
| 2001 | Hoffman | London | 100 | 1001 |
| 2002 | Giovanni | Rome | 200 | 1003 |
| 2003 | Liu | San Jose | 200 | 1002 |
| 2004 | Grass | Berlin | 300 | 1002 |
| 2006 | Clemens | London | 100 | 1001 |
| 2008 | Cisneros | San Jose | 300 | 1007 |
| 2007 | Pereira | Rome | 100 | 1004 |
+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

```
mysql> select * from orders;
```

| Onum | Amt     | Odate      | Cnum | Snum |
|------|---------|------------|------|------|
| 3001 | 18.69   | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19  | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75   | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-05 | 2006 | 1001 |
| 3010 | 1309.95 | 1990-10-06 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-06 | 2006 | 1001 |

```
10 rows in set (0.00 sec)
```

### Assignment –2

Introducing Relational Databases.

1) Which field of the Customers table is the primary key?

Answer: CNum column is the primary key of the Customers table;

2) What is the 4th column of the Customers table?

Answer: Rating column is the 4<sup>th</sup> column of the Customers table.

3) What is another word for row? For column?

Answer: Synonyms for :

Row = [Tuple, Record]

Column = [Field, Attribute]

4) Why isn't it possible to see the first five rows of a table?

Answer:

### Assignment –3

Overview of SQL

1) Does ANSI recognize the data type DATE?

Ans: Yes, ANSI does recognize data type DATE.

2) Which subdivision of SQL is used to insert values in tables?

Ans: For inserting values in tables we use "insert into" command which belongs to DML (Data Manipulation language) subdivision of SQL.

### Assignment –4

Retrieving Information from Tables.

1) Write a select command that produces the order number, amount, and date for all rows in the Orders table.

Answer:

```
mysql> select Onum,Amt,Odate from Orders;
```

```
mysql> select Onum,Amt,Odate from Orders;
+-----+-----+-----+
| Onum | Amt   | Odate   |
+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 |
| 3003 | 767.19 | 1990-10-03 |
| 3002 | 1900.10 | 1990-10-03 |
| 3005 | 5160.45 | 1990-10-03 |
| 3006 | 1098.16 | 1990-10-03 |
| 3009 | 1713.23 | 1990-10-04 |
| 3007 | 75.75 | 1990-10-04 |
| 3008 | 4723.00 | 1990-10-05 |
| 3010 | 1309.95 | 1990-10-06 |
| 3011 | 9891.88 | 1990-10-06 |
+-----+-----+-----+
10 rows in set (0.05 sec)
```

```
+-----+-----+-----+
```

2) Write a query that produces all rows from the Customers table for which the salesperson's number is 1001.

Answer: select \* from customers where Snum=1001;

```
mysql> select * from customers where Snum=1001;
+-----+-----+-----+-----+-----+
| Cnum | Cname | City   | Rating | Snum |
+-----+-----+-----+-----+-----+
| 2001 | Hoffman | London | 100    | 1001 |
| 2006 | Clemens | London | 100    | 1001 |
+-----+-----+-----+-----+-----+
2 rows in set (0.07 sec)
```

3) Write a query that displays the Salespeople table with the columns in the following order: city, sname, snum, comm.

Answer: mysql> select City,Snum,Sname,Comm from Salespeople;

```
mysql> select City,Snum,Sname,Comm from Salespeople;
+-----+-----+-----+-----+
| City   | Snum | Sname | Comm |
+-----+-----+-----+-----+
| London | 1001 | Peel  | 0.12 |
| San Jose | 1002 | Serres | 0.13 |
| London | 1004 | Motika | 0.12 |
| Barcelona | 1007 | Rifkin | 0.15 |
| New York | 1003 | Axelord | 0.10 |
+-----+-----+-----+-----+
5 rows in set (0.01 sec)
```

4) Write a select command that produces the rating followed by the name of each customer in San Jose.

Answer:

mysql> select Rating , Cname from customers where City='San Jose';

```
mysql> select Rating , Cname from customers where City='San Jose';
+-----+-----+
| Rating | Cname |
+-----+-----+
| 200    | Liu   |
| 300    | Cisneros |
+-----+-----+
2 rows in set (0.01 sec)
```

5) Write a query that will produce the snum values of all salespeople (suppress the duplicates) with orders in the Orders table.snum

Answer: select distinct snum from Orders;

```
mysql> select distinct snum from Orders;
+-----+
| snum |
+-----+
| 1007 |
| 1001 |
| 1004 |
| 1002 |
| 1003 |
+-----+
5 rows in set (0.01 sec)
```

## Assignment -5

### Relational and Logical

Operators.

1) Write a query that will give you all orders for more than Rs. 1,000.

mysql> select \* from Orders where Amt>=1000;

```
mysql> select * from Orders where Amt>=1000;
+-----+-----+-----+-----+-----+
| Onum | Amt   | Odate   | Cnum | Snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-05 | 2006 | 1001 |
| 3010 | 1309.95 | 1990-10-06 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-06 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

2) Write a query that will give you the names and cities of all salespeople in London with a commission above .10.

Answer: mysql> select Sname,City from Salespeople where city='London' AND Comm>.10;

```
mysql> select Sname,City from Salespeople where city='London' AND Comm>.10;
+-----+-----+
| Sname | City |
+-----+-----+
| Peel  | London |
| Motika | London |
+-----+-----+
2 rows in set (0.01 sec)
```

3) Write a query on the Customers table whose output will exclude all customers with a rating <= 100, unless they are located in Rome.

4) What will be the output from the following query?

Select \* from Orders

where (amt < 1000 OR

NOT (odate = '1990-10-03'

AND cnum > 2003));

```
mysql> Select * from Orders
-> where (amt < 1000 OR
-> NOT (odate = '1990-10-03'
-> AND Cnum > 2003));
+-----+-----+-----+-----+-----+
| Onum | Amt   | Odate   | Cnum | Snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-05 | 2006 | 1001 |
| 3010 | 1309.95 | 1990-10-06 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-06 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
8 rows in set (0.01 sec)
```

5) What will be the output of the following query?

Select \* from Orders

where NOT ((odate = '1990-10-03' OR snum

>1006) AND amt >= 1500);

```
mysql> Select * from Orders
-> where NOT((odate='1990-10-03' OR snum
-> >1006) AND amt>=1500);
```

| Onum | Amt     | Odate      | Cnum | Snum |
|------|---------|------------|------|------|
| 3001 | 18.69   | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19  | 1990-10-03 | 2001 | 1001 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75   | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-05 | 2006 | 1001 |
| 3010 | 1309.95 | 1990-10-06 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-06 | 2006 | 1001 |

8 rows in set (0.00 sec)

6) What is a simpler way to write this query?

Select snum, sname, city, comm From Salespeople

where (comm > .12 OR comm <.14);

```
mysql> Select snum,sname,city,comm From Salespeople
-> where comm Between .12 AND .14;
```

| snum | sname  | city     | comm |
|------|--------|----------|------|
| 1002 | Serres | San Jose | 0.13 |

1 row in set (0.01 sec)

```
mysql>
```